



Figure 2 – **Proposed secondary structure of the polymerase ribozyme QT45.** Schematic representation of the 45-nucleotide catalytic motif, read from the 5' to the 3' end, with numbering indicated at intervals of five residues. Each nucleotide is identified by its base (A, C, G, U). The dotted lines denote the Watson–Crick base pairings that define the double-helical stems, whereas the arrows indicate the connectivity and the directionality of the backbone. This compact architecture, notably smaller than that of the polymerases derived from the class I ligase, illustrates how the functions required for template-dependent RNA synthesis can be encoded in a motif of reduced size, a central point for the argument developed in Section [K.3.1](#) (GIANNI et al., 2026).

